

Jayaraman Venkatesan

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EXPERIENCE

United Wholesale Mortgage

Software Development Engineer - III

Pontiac, MI

Sep 2024 - Current

- **Built** a unified payment service (Stripe, .NET Core) processing 4–5K transactions/day (\$10M/month), with pluggable vendor abstraction enabling zero-downtime provider migration.
- **Designed** a C# edge auth gateway (OIDC, JWT) that validates users, mints a signed identity passport, and distributes it to 20+ downstream microservices via a shared SDK.
- **Developed** a .NET Core authorization SDK (Cosmos DB, RBAC/ACL) adopted across 12+ services, eliminating 2K+ lines of duplicated security logic.
- **Standardized** enterprise identity model (RBAC, ACLs) across 20+ services and 6+ teams, enabling consistent authorization and eliminating SSO vendor lock-in.
- **Built** and maintained a C# Kafka client library (Confluent SDK) adopted by 50+ teams across the organization, processing 50M+ events/day with built-in retries, DLQs, and scaling abstractions, increasing deploy frequency of event driven services by more than 50%.
- **Chaired** the Software Development Standards Council across 10+ teams, mandating WireMock integration testing for 20+ Tier-0/1 services and reducing change failure rate by 35%.
- **Modernized** a critical service from .NET Framework to .NET Core, containerized with Docker, and deployed to Kubernetes (AKS), decreasing change lead time by 70%, and decreasing mean time to recovery by 2hrs.
- **Built** a C# event-sourced loan export service (Kafka, .NET Core) serving 6500+ brokers in real time, exporting 20+ loan event types with sub-20s latency via configurable delivery channels.
- **Architected** a C# configuration-driven starter kit for creating new .NET Core microservices services along with complete deploy pipeline support, reducing team onboarding from 3 days to under 4 hours across 100+ teams.

Supply Trace, Northeastern University

Graduate Research Assistant

Boston, MA

Sept 2023 - May 2024

- **Built** a supply chain risk web app (ReactJS, Django, PostgreSQL) identifying forced labor risks across 100,000+ companies in a global shipment network via graph analysis (supplytrace.org).
- **Designed** Apache Airflow pipelines to clean, resolve, and geocode a 10M+ row dataset, enabling reliable downstream analytics.
- **Implemented** BFS, shortest-path, and connected-component algorithms using PostgreSQL recursive CTEs over 500K+ nodes and 2M+ edges, cutting risk propagation analysis from 4 hours to under 5 minutes.

Systems Research Group, Northeastern University

Graduate Research Assistant

Boston, MA

May 2023 - May 2024

- **Researched** isolation-level correctness across 10+ ACID databases, building a deterministic C# test framework to evaluate and classify concurrency violations.
- **Built** a SQL transaction simulation suite generating 1M+ interleaving transactions, exposing isolation-level bugs in MariaDB and Dgraph.
- **Mentored** 80+ graduate and PhD students as Teaching Assistant for computer systems courses, leading office hours on operating systems topics.
- **Contributed** to a research paper analyzing real-world concurrency anomalies across ACID-compliant databases, synthesizing findings from 500+ targeted test scenarios into a structured taxonomy of isolation-level violations.

Visa Inc
Software Development Engineer

Bengaluru, India
May 2020 - Jul 2022

- **Built** a global billing platform (Spring Boot, Angular) supporting 14,000+ financial institutions, contributing to 33% of Visa's revenue pipeline.
- **Developed** a batch processing microservice (Spring Batch, ActiveMQ) validating 1M+ metrics per run, saving 15+ hours/quarter across 100K+ banks.
- **Championed** mutation testing (PIT, Java) across 8+ services, raising coverage from 42% to 80%, gating risky deployments via Docker build hooks.
- **Built** a real-time analytics dashboard (Angular, TypeScript) aggregating 50+ KPIs for 28K+ employees, improving operational visibility enterprise-wide.
- **Optimized** data ingestion pipelines (Apache NiFi, Talend) processing 2TB+/day from 15+ sources into a centralized data lake, reducing latency by 55% and improving report availability by 3 hours.

Visa Inc
Software Development Engineer - Intern

Bengaluru, India
May 2019 - Jul 2019

- **Deployed** a recommendation engine (Apache Mahout, Java) using Google Analytics clickstream data, improving accuracy by 20% for 28K+ internal users.
- **Integrated** ELK stack (Elasticsearch, Logstash, Kibana) across 10+ services, cutting mean debug time by 60% and resolving critical issues in 15 min vs. 4+ hours previously.

EDUCATION

Northeastern University
Master of Science in Computer Science (**GPA: 4.0 / 4.0**)

Boston, MA
Sep 2022 - May 2024

Anna University
Bachelor of Engineering in Computer Science

Chennai, India
Aug 2016 - May 2020

SKILLS

Programming: C#, Java, Python, SQL, C++

Frameworks: .NET Core, Django, Spring Boot, ReactJS

Cloud: Microsoft Azure (AKS, Blob, Queues, ACL, Cosmos)

Tools: Kafka, Kubernetes, Docker, PubSub

Certified: Azure Certified Cloud Practitioner, Certified Kubernetes Application Developer

PROJECTS

EGOS-2K+ (Microkernel OS) [\(Code\)](#)

- Implemented a microkernel-based operating system (3K LOC) extending Cornell's EGOS-2000, adding threading libraries, file system support, virtual memory (page tables), and multilevel feedback queue scheduling.

Secure File Sharing in Cyber-Physical Systems [\(Report\)](#)

- Designed a secure file-sharing system using Elliptic Curve Cryptography, implementing asymmetric encryption protocols to ensure robust data confidentiality and integrity across distributed environments.

Smart Attendance System

- Built a real-time facial recognition system using Raspberry Pi and computer vision models, enabling biometric-based attendance tracking and presented at a government-sponsored tech expo.

Come Sell With Me [\(Backend Code\)](#) | [\(Frontend Code\)](#)

- Designed and developed an e-commerce platform using React for the frontend, Node.js and Express for the backend, and MongoDB for data management, with Azure Storage integration for image handling.